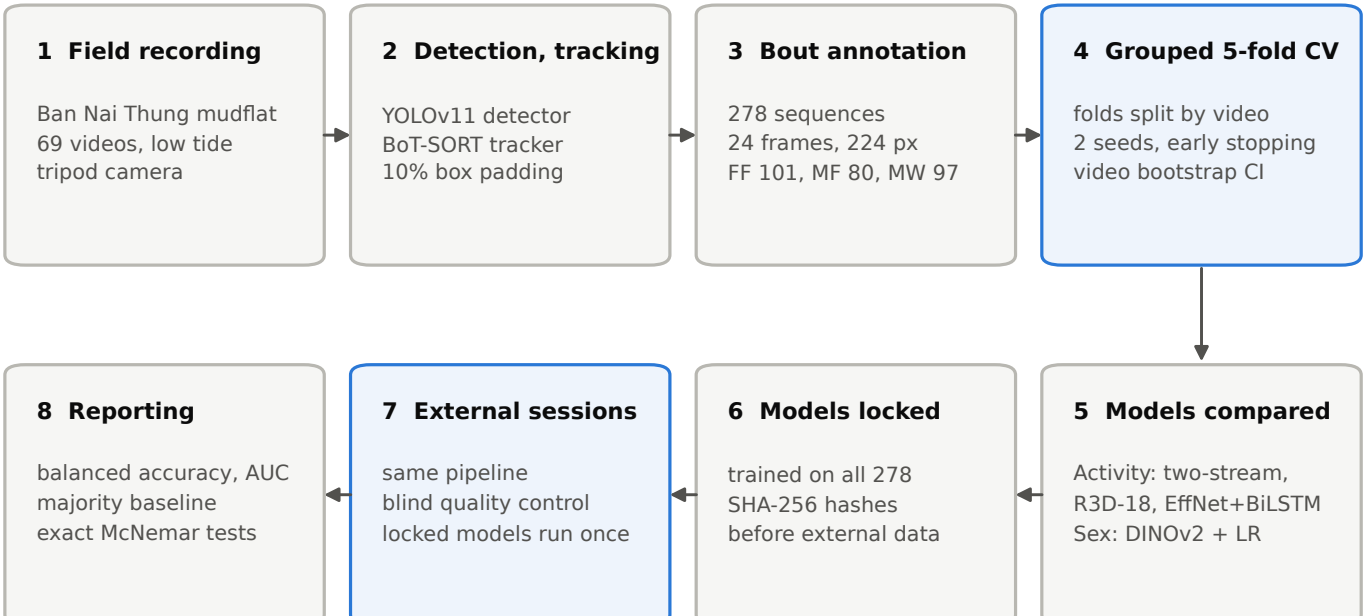
1 Field recording
Ban Nai Thung mudflat
69 videos, low tide
tripod camera
2 Detection, tracking
YOLOv11 detector
BoT-SORT tracker
10% box padding
3 Bout annotation
278 sequences
24 frames, 224 px
FF 101, MF 80, MW 97
4 Grouped 5-fold CV
folds split by video
2 seeds, early stopping
video bootstrap CI
5 Models compared
Activity: two-stream,
R3D-18, EffNet+BiLSTM
Sex: DINOv2 + LR
6 Models locked
trained on all 278
SHA-256 hashes
before external data
7 External sessions
same pipeline
blind quality control
locked models run once
8 Reporting
balanced accuracy, AUC
majority baseline
exact McNemar tests